

GoalWise — Critical Design Review

v0.2.0 Integrated Staging Release · Core MVP planning loop

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Goal: tell a user whether they are on pace for one near-term savings goal

Design stance: deterministic financial core, AI-free calculations, auditable snapshots

CDR focus: refined design, implementation progress, test strategy, risks, and final plan

CORE STACK



Preparation note: OpenAI Codex assisted with slide drafting, layout generation, and formatting; the team reviewed and owns the content.

Since the PDR

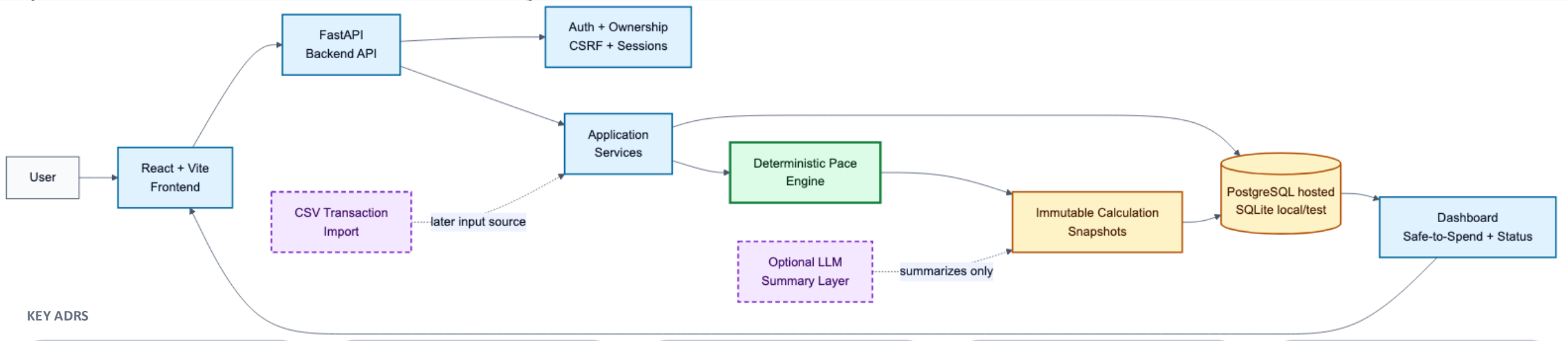
Visible changes from preliminary design to detailed design

Requirements	SRS v2.0 review + phased build path made explicit
Frontend	React/Vite workflow integrated with auth, goal, financial inputs, dashboard
Backend	Goal lifecycle, dashboard reads, snapshots, weekly plans, API conventions
Deployment	Railway staging + Caddy same-origin /api proxy for hosted browser behavior
Review	Quality gates + historical findings fed back into implementation work

CDR answer to PDR feedback
The project now has a buildable layered design, multiple accepted ADRs, a real integrated core, and a clear next path for CSV import and AI explain-only work.

Architecture & Key Decisions

Layered modular monolith with a deterministic financial core



ADR-0001 Layered modular monolith	ADR-0002 Deterministic engine; no runtime AI	ADR-0003 Immutable calculation snapshots	ADR-0005 Server sessions + ownership checks	ADR-0008/9 React/Vite + Railway deployment
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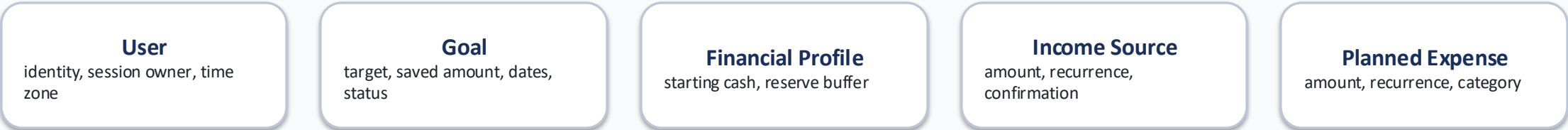
Defensible trade-off
Build CSV import and AI explanations next, but keep money guidance deterministic, testable, and backend-owned.

Detailed Design — Class / Module View

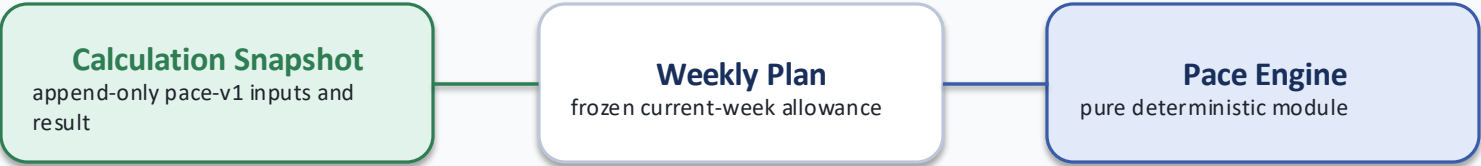
Buildable backend boundaries and core domain objects



CORE PERSISTED ENTITIES

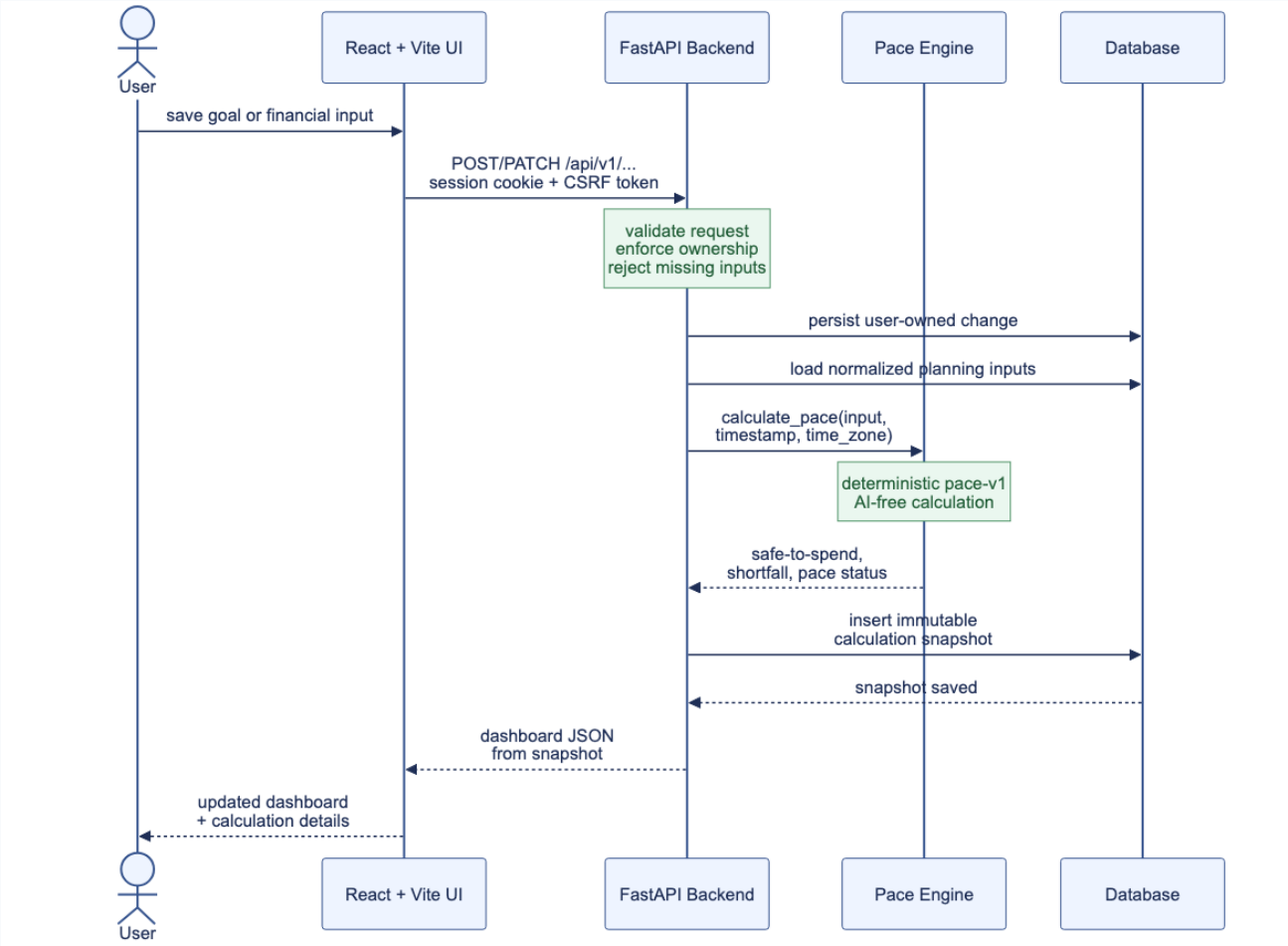


CALCULATION TRAIL



Detailed Design — Sequence

Save input → recalculate → snapshot → dashboard refresh



Key validation

Authenticated session
CSRF on unsafe methods
Server-side ownership checks

Key invariant

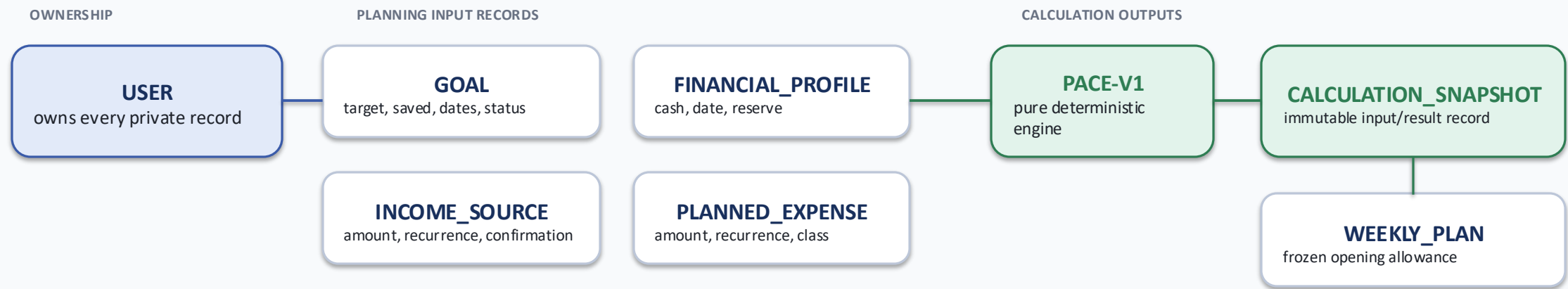
The frontend triggers changes, but the backend owns official calculations.

Traceability

Every valid change can create an immutable pace-v1 snapshot.

Detailed Design — Data Model

User-owned planning inputs produce immutable calculation history



DATA RULES

- Money stored as integer cents, never floats
- UTC timestamps; local-calendar semantics for date-only values
- Every protected row belongs to user_id
- Cross-user private resource access returns 404
- Snapshot rows are inserted, not mutated

Next demo increments

CSV import adds transaction inputs. AI summaries explain committed snapshot results. Neither replaces pace-v1 as the source of truth.

Implementation Progress + Code Review

v0.2.0 integrated staging release

Built now	Auth/session/CSRF, goal lifecycle, manual inputs, pace engine, snapshots, dashboard API, React workflow
Integrated	v0.2.0 hosted staging core; frontend uses same-origin /api proxy
Next	CSV import and AI explain-only layer for the upcoming demo increment
Reviewed	AI-assisted code review + manual SRS workflow checks + CI gates

REVIEW FINDINGS

High: hosted CSRF/cookie behavior

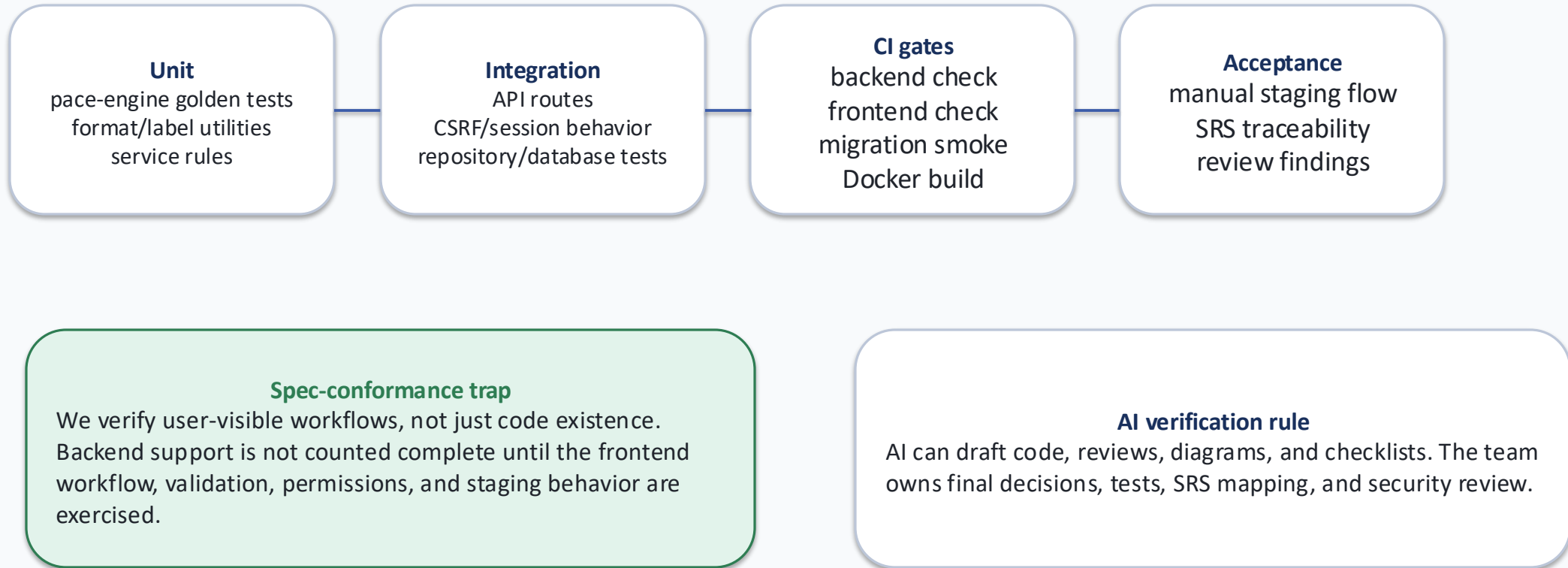
Resolved by Caddy same-origin /api proxy and Railway staging verification.

Medium: API existed before full workflow

Goal archive gap was treated as a spec-conformance issue, not merely a backend task.

Test & Verification Strategy

Guarding against the spec-conformance trap



Risks, Schedule, and Plan to Final

Active risk management for the final push

Scope creep	Add CSV and AI as bounded next increments without moving calculation out of pace-v1
Deployment drift	Use Railway staging, Caddy /api proxy, health checks, and env-var review
Calculation trust	Protect deterministic engine with golden tests, integer cents, snapshots, formula version
UX completeness	Streamline the UI, reduce friction, improve intuitive flow, and preserve demo path stability

REMAINING WORK

Finalize CDR feedback → streamline UI → security pass → broaden acceptance testing → prepare final demo and release notes.

Team status: on track, with CSV import and AI summaries positioned as demo increments on top of the verified core.